

PDF file name: Wiley-Early life stress and HPA axis

Original article word count: 8131

Extracted content word count: 3141 (38% of the original word count)

Title of the article: *Early life stress and HPA axis function independently predict adult depressive symptoms in metropolitan Cebu, Philippines*

The content extracted is as follows:

1 | INTRODUCTION

The durable effects of early life psychosocial stress exposure are also known to predict adult depression beyond the influence of genetic predispositions, as these patterns have been reported in twin studies (Kendler et al., 2000; Nelson et al., 2002).

Elucidating the possible underlying biological correlates, in diverse cultural contexts, can allow for the identification of universal biocultural pathways that contribute to the lasting impacts of early life experiences on adult depression and can illuminate potential opportunities to ameliorate the long-term poor health outcomes shaped by these adverse experiences and conditions.

The hypothalamic pituitary adrenal (HPA) axis, a primary physiological system involved in the mammalian stress response, has been implicated as an important biological system that sustains the impacts of early life psychosocial and environmental exposures and shapes later life mental health status (Heim & Binder, 2012; Miller, Chen, & Parker, 2011; Taylor, Way, & Seeman, 2011).

Early life stress exposure and adverse social, emotional, and physical conditions have consistently been reported as strong predictors of HPA axis dysregulation as indexed by abnormal diurnal cortisol rhythms and reactions to stressful events (Gustafsson, Janlert, Theorell, & Hammarström, 2010; Heim & Binder, 2012; Taylor et al., 2011; Thayer, Wilson, Kim, & Jaeggi, 2018), though the direction of these relationships has varied across the literature.

For example, individuals who experienced early life stress have exhibited both higher (Luecken & Appelhans, 2006; Schalinski, Elbert, Steudte-Schmiedgen, & Kirschbaum, 2015; van der Vegt, Van Der Ende, Kirschbaum, Verhulst, & Tiemeier, 2009) and lower levels of basal cortisol (DeSantis, Kuzawa, & Adam, 2015; Trickett, Noll, Susman, Shenk, & Putnam, 2010; van der Vegt et al., 2009) during adulthood.

Early life stress is also related to an altered cortisol awakening response (Butler, Klaus, Edwards, & Pennington, 2017; Engert, Efanov, Dedovic, Dagher, & Pruessner, 2011; Fogelman & Canli, 2018; Gonzalez, Jenkins, Steiner, & Fleming, 2009) and evening cortisol levels (Engert et al., 2011; Gustafsson et al., 2010) in adulthood in both positive and negative directions.

Additionally, adults with a history of child abuse and neglect have been reported to exhibit signs of HPA axis dysregulation as evidenced by greater (Heim et al., 2000; Luecken & Appelhans, 2006; Pesonen et al., 2010; Vaccarino, Levitan, & Ravindran, 2015) and blunted levels (Carnut et al., 2015; Cris an, Vulturar, Opre, & Miu, 2015; Carpenter et al., 2007; Carpenter, Tell, Gaylord-Harden, & Mathews, 2017; Lovallo, Farag, Sorocco, Cohoon, & Vincent, 2012) of cortisol reactivity in response to acute psychosocial stressors.

Early life stress-induced HPA-axis dysregulation has also been seen in non-human primates.

Thus, prior research has emphasized that alterations in HPA axis function are not only affected by the embodiment of early life and recent social experiences but may also predict individual risk for depression later in life.

The role of the HPA axis in shaping the depressive effects of early life stress, however, remains unclear as results from past studies have reported mixed findings (Engert et al., 2011; Gerritsen et al., 2010; Gonzalez et al., 2009; Mangold et al., 2011; Zilioli et al., 2016).

Although growing evidence has implicated early life stress-induced alterations in HPA axis function as a possible mechanism for adult depression, the potential psychobiological mechanisms that underlie the long-term depressive effects of early life stress are not wellknown.

Additionally, these studies have mostly been conducted among Western and industrialized populations, where cultural, economic, and psychosocial environments vary in comparison to low- and middle-income contexts, where the burden of mental illness is the contexts and populations is necessary to assess the generalizability of the effects of early life social experience on downstream health outcomes and to identify the factors that may increase psychopathological risk.

Additionally, few studies explore the durability of the effects of early life conditions on HPA axis function, as a majority of studies tend to examine early life stress retrospectively, rather than prospectively.

Our aim in the current study is to explore whether the role of early life stress as a predictor of adult depression is accounted for by alterations in adult HPA axis function in a non-clinical, naturalistic setting.

We build on this work to test the hypothesis that a composite measure of early life stressors will predict depressive symptoms during early adulthood.

Additionally, we further hypothesize that altered HPA axis function, as reflected by elevated salivary bedtime cortisol and cortisol awakening response levels, will account for the depressive effects of early life stress.

2 | METHODS

Six measures of early life family-related stress were operationalized as dichotomous categorical variables and summed to create a composite score.

Early life stress variables included measures of maternal absence, paternal instability, sibling death, witnessing domestic violence during childhood (Gettler et al 2015; Hindin & Gultiano, 2006), and reverse scored measures of feeling close to one's mother during childhood, and closeness to one's father during childhood (Hindin, 2005).

These six variables were among the small handful of data points on early life stress that were collected during the study, which was initially designed to study the long-term health impacts of infant feeding.

Food consumption and exercise were also tested as covariates but were not included in the final models since they were not significantly related to early life stress, depressive symptoms, or individual cortisol indices.

In order to capture the overall SES environment during the timespan in which the experiences of early life stress occurred, between infancy and middle childhood, an early life SES variable was constructed by summing SES scores from the 1983, 1986, 1991, and 1994 surveys.

First, bivariate associations between early life stress, cortisol measures, depressive symptom scores, and covariates were calculated (Tables 2a and 2b).

Our final model examined whether or not associations between early life stress and depressive symptoms are dependent on our HPA axis measures by including measures of PM cortisol and the cortisol awakening response, then diurnal slope and the cortisol awakening response.

3 | RESULTS

Bivariate associations between major study variables are presented in Table 2a. Table 2b shows zero-order correlations between disaggregated early life stress measures and cortisol indices.

Our first model (Model 1) assessed the relationship between our early life stress composite measure and adult depressive symptoms controlling for sex (Table 3a).

Greater number of early life stressors significantly predicted a higher levels of depressive symptoms at age 21-22.

Results from this partial regression model also show a dose response relationship between the number of early life stressors and adult depressive symptoms.

We then controlled for possible effects attributed to socioeconomic status at both early life and 2005 (Model 2) by adding the multidimensional summary measure for both timepoints of SES.

We tested the relationship between adult depressive symptoms and diurnal cortisol slope (the rate of change in cortisol concentrations from morning to evening) and the cortisol awakening response while controlling for time of evening saliva collection, smoking status, as well as recent stress, which was assessed by an adapted version of the Perceived Stress Scale (Cohen et al., 1994).

Including the cortisol variables modestly altered the relationship between early life stress and depressive symptoms as indicated by the change in the coefficient of the early life stress composite variable from Model 2 (0.4006) to Model 4 (.353), the fully adjusted model, and the dummy coded stress groups.

Early life stress continues to significantly predict adult depressive symptoms in a dose response fashion.

Similar to the effects of diurnal slope and CAR variables in Models 3 and 4, the inclusion of the bedtime and waking cortisol variables led to very modest changes in the coefficient of the early life stress composite variable from Model 2 (0.4006) to Model 6 (0.39), the complete model, and the dummy coded stress groups.

We then disaggregated the early life stress composite measure to examine the effects of each early life stress variable, all of which were coded as categorical variables, on young adult depressive symptoms (Table 3b).

No other early life stress measures significantly predicted depression risk.

Being male and greater socioeconomic status at 2005, but not during early life, corresponded with decreased depressive symptom severity.

Finally, we conducted a separate analysis to examine whether early life stress affected young adult cortisol rhythms.

Our results show no significant relationships between the early life stress composite measure and any of the diurnal cortisol measures (e.g., waking, cortisol awakening response, evening, diurnal slope).

After disaggregating the early life stress composite measure, however, those who witnessed domestic violence during childhood exhibited significantly lower levels of bedtime cortisol ($p = .018$).

4 | DISCUSSION

Our findings suggest that early life stress, witnessing domestic violence during childhood, and alterations in adult HPA axis function predict the severity of depressive symptoms during adulthood in this sample of Cebuano young adults.

However, contrary to previous findings in the literature, these relationships appeared to be independent of one another, as indicated by the modest changes in the relationship between early life stress and adult depression risk after controlling for HPA axis measures (Tables 3a and 3b).

Thus, we report that diurnal cortisol activity does not affect adult depressive symptoms as a result of early life stress during child development.

Our early life stress composite measure also did not predict diurnal cortisol patterns during young adulthood in this sample, but witnessing domestic violence during childhood predicted lower bedtime cortisol levels.

Our findings linking early life stress and diurnal cortisol rhythms with depression are in general agreement with past work investigating similar questions.

We confirm the overall positive association previously reported between early life stress and adult depressive symptoms (Heim & Binder, 2012; Taylor et al., 2011) as seen by the direct relationship between our composite measure of early life stress and adult depressive symptoms (Figure 3) as well as the long-term adult depressive impacts of witnessing domestic violence during childhood.

When examining the relationships between each individual, disaggregated stress measure and young adult depression risk, with exception to witnessing domestic violence, none of our additional measures of early life stress significantly predicted depressive symptomatology independently (Table 3b).

The association between witnessing domestic violence and lower bedtime cortisol levels was an unexpected finding as past studies have generally found that early life adversity corresponds with greater evening levels.

Additionally, in a meta-analysis of chronic stress and HPA axis, experiencing chronic stressors that posed physical threats, were traumatic in nature, and uncontrollable were associated with elevated evening cortisol levels (Miller, Chen, & Zhou, 2007).

Although several studies have reported null associations between early life stress and adult evening cortisol concentrations (Gerritsen et al., 2010; Gonzalez et al., 2009; Miller et al., 2009; van der Vegt et al., 2009), a smaller number of studies have reported similar links between greater early adversity and lower evening cortisol levels that we report.

In our study, the association between decreased evening cortisol levels during young adulthood and witnessing domestic violence during adolescence may be a function of the timing since the onset of stress.

In the same meta-analysis, Miller et al. (2007) argue that the seemingly contradictory findings between hyper- and hypocortisolism among chronically-stressed adults simply reflect different timepoints during the stress process (35).

Upon exposure to stress, the HPA axis may become activated, causing a rise in circulating cortisol levels.

Additionally, studies on the early life impacts on the cortisol awakening response tend to report positive relationships.

For example, individuals exposed to an array of stressful life experiences during child development such as parental loss and child maltreatment (Gonzalez et al., 2009) and neglect and abuse (van der Vegt et al., 2009), and adverse social conditions such as low socioeconomic status (Gustafsson et al., 2010) generally exhibited heightened cortisol awakening responses compared to their respective comparison groups.

Similar to our prior consideration with depression, differences in the early life stress-HPA axis relationship seen across these studies may be explained by the particular qualities of early life stress in question, such as the frequency, severity, and social meaning of the psychosocial stressor or environmental condition (Fogelman & Canli, 2018; Lyons & Parker, 2007; Miller et al., 2007).

This study further emphasizes the role that early life experiences have on shaping the HPA axis in this sample.

The direction of the relationships between both measures of early life adversity and depression, however, were opposing. Cumulative economic strain from prenatal development to early adulthood was associated with higher bedtime cortisol levels, lower CARs, and other characteristics of HPA axis dysregulation (i.e., lower total cortisol output, flatter cortisol slopes) while our findings show that witnessing domestic violence predicted lower bedtime cortisol levels.

As previously mentioned, the nature of the stressor in question may explain the varying relationships between early life stress and HPA axis function seen across studies, though the lack of specificity in our survey question on witnessing domestic violence preclude us from ascertaining the particular qualities of the stressor that may explain our discrepant finding.

Considering the inverse relationship between witnessing domestic violence and adult evening cortisol levels, the finding that bedtime cortisol levels do not influence the relationship between early life stress and depressive symptoms is expected statistically but conceptually perplexing.

The strengthened relationship between early life stress and adult depressive symptoms after including the bedtime cortisol variable suggests that early life stress predicts depressive symptoms despite the fact that individuals in our sample tend to exhibit cortisol levels of non-depressed individuals.

We intended on conducting a mediation analysis to evaluate the links between early life stress, cortisol levels, and depressive risk, but our results did not meet the criteria for mediation analysis as all three variables were not significantly correlated to one another.

While cross-sectional measure of HPA axis function may not be related to the depressive effects of early life stress during young adulthood, HPA axis function at earlier stages of development may still influence one's risk of later life depression (Adam et al., 2010; Bockting et al., 2012; Goodyer, Bacon, Ban, Croudace, & Herbert, 2009; Vrshek-Schallhorn et al., 2013).

These findings also suggest that additional biological mechanisms may facilitate the durable effects of early life experiences on later life health.

Other biological pathways of HPA axis function that have previously been implicated in the early life stress-adult depression pathway include upstream mechanisms like epigenetic variation (e.g., DNA methylation, histone modification) (Heim & Binder, 2012; Khulan et al., 2014; Smith, Parets, & Kim, 2014) and downstream pathways such as inflammatory (immune activation, gut-brain axis) (Pace et al., 2006) and neurobiological mechanisms (e.g., neurotransmission, neuroanatomic changes) (Gatt et al., 2010; Grant,

Cannistraci, Hollon, Gore, & Shelton, 2011).

For example, early life stress has been hypothesized to influence differential methylation signatures among HPA axis-related genes, such as the glucocorticoid receptor gene, which in may alter the binding of cortisol and its subsequent impact on adult depression (Heim & Binder, 2012; Jawahar, Murgatroyd, Harrison, & Baune, 2015).

Future studies can further elucidate the role of HPA axis function in contributing to the long-term effects of early life stress on adult depressive symptomatology by waiting to evaluate adult depressive risk a few months or years after cortisol measurements are collected. This longitudinal design can help clarify whether the durability of altered adult cortisol activity due to early life stress is what influences later life depressive symptoms and eliminates the possibility of reverse causality between cross-sectional measures of depressive symptoms and cortisol levels, as seen in our study.

Reliable and culturally validated measures of early life stress and depression will also strengthen the internal validity of future studies, and multiple days of cortisol collection with daily assessments of mood, behaviors, and experience will better characterize typical diurnal cortisol rhythms.

Finally, additional stress-sensitive mechanism that are both upstream and downstream of HPA axis function can be assessed to identify other biological pathways involved in the long-term depressive effects of early life stress.

The first limitation of this study is our crude measure of early life stress.

There are likely a wide variety of stressors across multiple domains and levels of life that may have similar long-term effects on mental health and, in general, affect the body.

Different dimensions of psychosocial stress, such as a more diverse set of experiences and conditions (various forms of social oppression, disease and illness experience, teratogens, etc.), chronicity, severity, and sociocultural meaning, could have distinct and varied sequelae on HPA axis function (e.g., diurnal cortisol rhythm, stress reactivity) and later life health.

5 | CONCLUSION

In summary, early life family-related stressors predicted greater severity of depressive symptoms at early adulthood in a dose response fashion and witnessing domestic violence during child development predicted lower levels of evening cortisol in this large, naturalistic, non-Western, and longitudinal sample of individuals in metropolitan Cebu, Philippines.

When considering the cumulative effects of early life family-based adversity, greater exposure to these stressors during development predicted worse depressive outcomes in early adulthood.

While both bedtime cortisol and diurnal cortisol slopes significantly predicted adult depressive symptoms cross-sectionally, adult cortisol levels did not explain the effect of early life stress on adult depressive symptoms.

Furthermore, this analysis does not support the notion that the depressive impacts of early life stress are influenced by HPA axis function in this particular stage of young adulthood and this sample.

Additionally, we also note that our early life stress composite variable is limited as it is not a validated measure of family-based stress in Cebu.

Nonetheless, these findings suggest that the long-term depressive effects of certain forms of early life stress extend to this large sample in the Philippines, and that early life stress and HPA axis function at age 21 22 in this sample may shape adult depression through independent pathways in our sample.

Our results provide further evidence that HPA axis activity is shaped by early life conditions and is associated with depressive symptoms, and that early life development is an important period research and intervention for improving global mental health.

Elevated cortisol awakening response associated with early life stress and impaired executive function in healthy adult males.

Emotional non-acceptance links early life stress and blunted cortisol reactivity to social threat.

Decreased adrenocorticotrophic hormone and cortisol responses to stress in healthy adults reporting significant childhood maltreatment.

Effect of childhood physical abuse on cortisol stress response.

The role of life stress and daily negative emotion.

Daytime cortisol and stress reactivity in the offspring of parents with bipolar disorder.

Diminished cortisol responses to psychosocial stress associated with lifetime adverse events: A study among healthy young subjects.

Increased cortisol awakening response and afternoon/evening cortisol output in healthy young adults with low early life parental care.

Early life stress combined with serotonin 3A receptor and brain-derived neurotrophic factor valine 66 to methionine genotypes impacts emotional brain and arousal correlates of risk for depression.

The relation between early life adversity, cortisol awakening response and diurnal salivary cortisol levels in postpartum women.

Current research trends in early life stress and depression: Review of human studies on sensitive periods, gene environment interactions, and epigenetics.

Lifetime adversity leads to blunted stress axis reactivity: Studies from the Oklahoma family health patterns project.

Parental caring and loss during childhood and adult cortisol responses to stress.